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Prompt Engineering with a Llama Model

The Llama 3.2 collection consists of multilingual large language models in 1B and 3B parameter sizes, designed for text-based input and output. These models include both pre-trained and instruction-tuned variants, explicitly optimized for multilingual dialogue applications such as agent-driven retrieval and summarization. They perform superior to many open-source and proprietary chat models on widely used industry benchmarks.

We will use the Llama-3.2-1B-Instruct from ♾ **Hugging Face meta-llama** space and use it offline.

Meta requires permission to download the model, which requires an `edu` email address. Follow the instructions to request permission. Then create an access token in 🤗 **your account** webpage. Ensure you enable the Repositories - Read access to the contents of all public gated repos you can access, so that your access token can enable the Meta models.

Set the `HF_HOME` as in Cell 1 so the model can be downloaded and used offline. Then, **login to Hugging Face** as in Cell 3 below by providing the access token in the prompt. If you installed 🐙 **Git** version control system, then the token will be stored automatically.

Note that the `temperature` hyperparameter has minimal effect in this model, unlike the larger gpt-3.5-turbo model we tried in the other Jupyter notebook.

The first two examples are basic prompts and changing the prompt/response temperature. Next are various examples, including few-shot prompts, CoT, zero-shot CoT, etc.

In [1]:
```
%matplotlib inline
import matplotlib.pyplot as plt
plt.rcParams['figure.dpi'] = 72
from IPython.display import Markdown
import numpy as np
import os

MODEL_PATH= '/EP_models/'
os.environ['HF_HOME'] = MODEL_PATH  # before import transformers
os.environ['HF_DATASETS_OFFLINE']= '1'
```

In [2]:
```
import torch
import transformers
from transformers import pipeline

# filter warnings
import warnings
transformers.logging.set_verbosity_error()
warnings.simplefilter(action='ignore', category=FutureWarning)
warnings.simplefilter(action='ignore', category=UserWarning)

Device = torch.device("cuda:0" if torch.cuda.is_available() else "cpu")

print(f'PyTorch version= {torch.__version__}')
print(f'transformers version= {transformers.__version__}')
print(f'CUDA available= {torch.cuda.is_available()}')
```

```
PyTorch version= 2.9.1+cu130
transformers version= 4.57.6
CUDA available= True
```

In [3]:
```
# from huggingface_hub import login
# login()
```

In [4]:
```
Model = pipeline("text-generation", model="meta-llama/Llama-3.2-1B-Instruc
```

In [5]:
```
def query(_model, _prompt):
    messages = [{'role':'user', 'content':_prompt}]
    response = _model(messages)
    return response[0]['generated_text'][1]['content']
```

In [6]:
```
Markdown(query(Model, 'Who are you?'))
```

Out[6]: I'm an artificial intelligence model known as Llama. Llama stands for "Large Language Model Meta AI."

In [7]:
```
Markdown(query(Model, 'The sky is ...'))
```

Out[7]: ... blue today.

In [8]:
```
Markdown(query(Model, 'be creative from now on. the sky is...'))
```

Out[8]: ...a deep, rich shade of indigo that stretches as far as the eye can see, punctuated by wispy strands of silver mist that dance and swirl like a thousand ethereal ballerinas. The air is filled with the sweet scent of blooming moonflowers, their delicate petals unfolding like tiny, shimmering lanterns in the fading light.

In [9]:
```
Markdown(query(Model, 'The sky is...'))
```

Out[9]: ...blue.

In [10]:
```
Markdown(query(Model, 'Please be creative and the sky is...'))
```

Out[10]: ...a canvas of swirling aurora borealis, with hues of emerald green, sapphire blue, and amethyst purple dancing across the velvet expanse. The stars twinkle like diamonds scattered across the sky, as the moon casts a silvery glow, illuminating the ethereal beauty of the celestial ballet.

A delicate mist creeps across the horizon, carrying the whispers of ancient myths and legends. The wind whispers secrets to the trees, and the leaves rustle in an otherworldly language, their soft susurrations weaving a spell of enchantment.

In the distance, a majestic mountain range stretches towards the heavens, its peaks shrouded in a mystical veil of fog and mist. The air is crisp and clean, filled with the sweet scent of blooming wildflowers, their petals swaying gently in the breeze like ballerinas dancing across the sky.

As the sun begins to set, the colors of the sky deepen and richen, a breathtaking display of pinks, oranges, and purples that seems to pulse with an inner light. The stars twinkle like diamonds scattered across the velvet expanse, and the moon rises high in the sky, casting a silver glow over the landscape.

The world is bathed in an ethereal light, a magical realm where dreams and reality blend together in a swirling vortex of color and sound. It's a place where the imagination knows no bounds, and the possibilities are endless.

In [11]:
```
# text summarization
prompt = """Antibiotics are a type of medication used to treat bacterial i
They work by either killing the bacteria or preventing them from reproduci
Antibiotics are usually taken orally in the form of pills, capsules, or li
They are not effective against viral infections, and using them inappropri

Explain the above in one short sentence:"""

Markdown(query(Model, prompt))
```

Out[11]: Antibiotics are medications used to treat bacterial infections by either killing bacteria or preventing their reproduction.

In [12]:
```
# question answering
prompt = """Answer the question based on the context below. Keep the answe
Respond "Unsure about answer" if not sure about the answer.

Context: Teplizumab traces its roots to a New Jersey drug company called O
There, scientists generated an early version of the antibody, dubbed OKT3.
Originally sourced from mice, the molecule was able to bind to the surface
In 1986, it was approved to help prevent organ rejection after kidney tran

Question: What was OKT3 originally sourced from?
```

```
Answer:"""

Markdown(query(Model, prompt))
```

Out[12]: Unsure about answer

In [13]:
```
# text classification
prompt = """Classify the text into neutral, negative or positive.

Text: I think the food was okay.

Sentiment:"""

Markdown(query(Model, prompt))
```

Out[13]: To classify the text, I would say:

- Neutral: The text expresses a neutral sentiment, indicating no strong emotions or opinions.
- Negative: The word "okay" is used, which has a slightly negative connotation, implying the food was not exceptional or outstanding.
- Positive: The word "okay" is also used, which has a slightly positive connotation, implying the food was not bad or unpleasant.

In [14]:
```
# role playing
prompt = """The following is a conversation with an AI research assistant.

Human: Hello, who are you?
AI: Greeting! I am an AI research assistant. How can I help you today?
Human: Can you tell me about the creation of blackholes?
AI:"""

Markdown(query(Model, prompt))
```

Out[14]: Human: I'd like to inquire about the astrophysical phenomenon of black holes. Can you provide an overview of their formation and characteristics? AI: Black holes are regions in spacetime where the gravitational pull is so strong that nothing, not even light, can escape. Their creation is attributed to the collapse of massive stars, particularly those with masses in excess of 3-4 times the solar mass. The process involves the following steps:

1. A massive star undergoes a supernova explosion, leaving behind a core of heavy elements.
2. If the core is massive enough, it undergoes a process known as core collapse, where the gravitational force overcomes the nuclear binding energy, causing a massive release of energy.
3. The resulting singularity is a point of infinite density and zero volume, with a surface known as the event horizon.
4. The event horizon marks the boundary beyond which anything that enters cannot escape, including light.

The event horizon is not a physical boundary but rather a mathematical concept, where the curvature of spacetime becomes so extreme that the laws of physics as we know them break down. Once matter crosses the event horizon, it is effectively trapped by the black hole, and any attempt to escape is futile.

AI: I'd like to highlight that the concept of black holes is deeply rooted in Einstein's general theory of relativity, which postulates that spacetime is curved by the presence of mass and energy. The curvature of spacetime around a massive object such as a black hole is what gives rise to the phenomenon of gravitational lensing and the bending of light around massive objects.

In [15]:
```
# code generation
prompt = """
"\"\"\"\nTable departments, columns = [DepartmentId, DepartmentName]\nTabl
columns = [DepartmentId, StudentId, StudentName]\nCreate a MySQL query for
"""

Markdown(query(Model, prompt))
```

Out[15]: To create a MySQL query that retrieves all students in the Computer Science Department, you can use the following query:

```
SELECT *
FROM students
WHERE DepartmentId = 2;
```

In this query:

- `DepartmentId = 2` specifies that we want to retrieve students from the Computer Science Department (ID 2).
- `SELECT *` selects all columns from the `students` table.
- `FROM students` specifies the table from which to retrieve data.
- `WHERE DepartmentId = 2` filters the data to only include students from the specified department.

Note: The ID of the Computer Science Department is assumed to be 2. If your department IDs are different, you should replace 2 with the correct ID.

If you want to retrieve all students, including those from other departments, you can use a `UNION` operator to combine the results of two separate queries:

```
SELECT *
FROM students
WHERE DepartmentId = 1;
SELECT *
FROM students
WHERE DepartmentId = 2;
```

In this query:

- `SELECT * FROM students WHERE DepartmentId = 1` retrieves students from the Sales Department.
- `SELECT * FROM students WHERE DepartmentId = 2` retrieves students from the Computer Science Department.
- `UNION` combines the results of these two queries into a single result set.

In [16]:

```
# reasoning
prompt = """The odd numbers in this group add up to an even number: 15, 32

Solve by breaking the problem into steps. First, identify the odd numbers,

Markdown(query(Model, prompt))
```

Out[16]: To solve the problem, we will follow these steps:

1. Identify the odd numbers in the group: The odd numbers in the group are 15, 5, 13, 7, and 1.

2. Add the odd numbers: 15 + 5 = 20 20 + 13 = 33 33 + 7 = 40 40 + 1 = 41

3. Determine if the result is odd or even: The sum of the odd numbers (41) is an odd number.

Therefore, the statement "The odd numbers in this group add up to an even number" is false.

In [17]:
```
# few shot prompts
prompt = """The odd numbers in this group add up to an even number: 4, 8,
A: The answer is False.

The odd numbers in this group add up to an even number: 17,  10, 19, 4, 8,
A: The answer is True.

The odd numbers in this group add up to an even number: 16,  11, 14, 4, 8,
A: The answer is True.

The odd numbers in this group add up to an even number: 17,  9, 10, 12, 13
A: The answer is False.

The odd numbers in this group add up to an even number: 15, 32, 5, 13, 82,
A: """

Markdown(query(Model, prompt))
```

Out[17]: Let's analyze the given groups one by one:

A: The odd numbers in this group add up to an even number: 17, 10, 19, 4, 8, 12, 24. The sum of these odd numbers is 17 + 10 + 19 + 4 + 8 + 12 + 24 = 94. Since 94 is an even number, the statement is True.

B: The odd numbers in this group add up to an even number: 16, 11, 14, 4, 8, 13, 24. The sum of these odd numbers is 16 + 11 + 14 + 4 + 8 + 13 + 24 = 86. Since 86 is an even number, the statement is True.

C: The odd numbers in this group add up to an even number: 17, 9, 10, 12, 13, 4, 2. The sum of these odd numbers is 17 + 9 + 10 + 12 + 13 + 4 + 2 = 65. Since 65 is an odd number, the statement is False.

Therefore, the correct answer is:

A: The answer is True.

In [18]:
```
# chain-of-thought
prompt = """The odd numbers in this group add up to an even number: 4, 8,
A: Adding all the odd numbers (9, 15, 1) gives 25. The answer is False.

The odd numbers in this group add up to an even number: 15, 32, 5, 13, 82,
A:"""

Markdown(query(Model, prompt))
```

Out[18]: To solve this, let's first identify the odd numbers in the group: 9, 15, 1, 13, 7.

Now, let's add these odd numbers: 9 + 15 = 24 24 + 1 = 25 25 + 13 = 38 38 + 7 = 45

The sum of these odd numbers is 45, which is an odd number. This means the statement is False.

In [19]:
```
# zero shot CoT
prompt = """I went to the market and bought 10 apples.
I gave 2 apples to the neighbor and 2 to the repairman.
I then went and bought 5 more apples and ate 1.
How many apples did I remain with?

Let's think step by step."""

Markdown(query(Model, prompt))
```

Out[19]: To find out how many apples you have left, let's break down the steps:

1. You initially bought 10 apples.
2. You gave 2 apples to the neighbor, leaving you with:
   - 10 (initial) - 2 (given away) = 8 apples
3. You gave 2 more apples to the repairman, leaving you with:
   - 8 (apples remaining after giving to neighbor) - 2 (given to repairman) = 6 apples
4. You then bought 5 more apples, but you ate 1. So, subtract the apples you ate:
   - 6 (apples remaining after giving to repairman) - 1 (eaten) = 5 apples

You have 5 apples remaining.

In [20]:
```
# self-consistency
question = "What are some good activities to relax after a long day?"

# generate multiple responses for self-consistency
responses = []
for _ in range(5):  # 5 responses for consistency
    messages = [{"role": "user", "content": question}]
    response = Model(messages)[0]['generated_text'][1]['content']
    responses.append(response)
```

```python
for idx, resp in enumerate(responses, start=1):
    print(f"Response {idx}: {resp}")

# select the most frequent answer
from collections import Counter
most_common_answer = Counter(responses).most_common(1)[0][0]

print("\n  ** The most consistent answer: ", most_common_answer)
```

Response 1: Relaxing after a long day can be a great way to unwind and recha rge. Here are some activities that can help you relax:

1. **Reading**: Get lost in a book, whether it's a fiction novel, a self-hel p book, or a magazine. Reading can be a great way to escape reality and rela x.

2. **Listening to music**: Put on your favorite playlist or find a relaxing music station on streaming services. Singing along or simply enjoying the ca lming sounds can be soothing.

3. **Taking a warm bath or shower**: Fill your bathtub with warm water, add some bubbles or essential oils, and soak for a while. Alternatively, take a relaxing shower to help your muscles unwind.

4. **Meditation and yoga**: Practice mindfulness meditation, yoga, or guided relaxation exercises to calm your mind and body. You can find many free reso urces online.

5. **Taking a walk**: Get some fresh air and take a short walk around your n eighborhood or a nearby park. Nature can be a great stress-reliever.

6. **Watching a movie or TV show**: Pick a favorite film or series and spend some time binge-watching. Just be sure to take breaks and relax before getti ng back to the screen.

7. **Playing with a pet**: If you have a furry friend at home, spend some ti me playing with them. Interacting with animals can be calming and stress-red ucing.

8. **Cooking or baking**: Prepare a simple meal or bake some cookies. Cookin g can be a fun way to relax and express your creativity.

9. **Doing some light stretching**: Take a few minutes to stretch your body, focusing on your major muscle groups. This can help release tension and rela x your muscles.

10. **Practicing deep breathing**: Take slow, deep breaths, focusing on the sensation of the air moving in and out of your body. This can help calm your mind and reduce stress.

11. **Sipping herbal tea**: Try a calming herbal tea, such as chamomile or l avender, to help you relax and unwind.

12. **Playing a board game or card game**: Spend some time playing a low-key game, like solitaire or a puzzle, to relax and have fun.

13. **Writing in a journal**: Write down your thoughts, feelings, and experi ences. Reflecting on your day can help you process your emotions and relax.

14. **Getting a massage**: Treat yourself to a professional massage or ask a partner or friend to give you a massage at home. This can be a great way to relax and unwind.

15. **Taking a power nap**: If you're feeling extremely tired, take a short nap (20-30 minutes) to recharge and refresh your mind.

Remember, everyone is different, so experiment with different activities to find what works best for you and relaxes you.
Response 2: There are many great activities to help you relax after a long day. Here are some suggestions:

1. **Take a warm bath or shower**: Fill your bathtub with warm water and add some calming essential oils like lavender or chamomile. Alternatively, take a relaxing shower with warm water and some gentle soap.

2. **Listen to soothing music**: Put on your favorite relaxing music or find a calming playlist on streaming services. You can also try listening to nature sounds, like rain or ocean waves.

3. **Practice gentle stretches or yoga**: Engage in some gentle stretches or follow a simple yoga routine to help release tension and relax your muscles.

4. **Read a book or magazine**: Get lost in a good book or catch up on a favorite magazine. This can help take your mind off your daily worries and transport you to a different world.

5. **Take a nap**: Sometimes, all you need is a short nap to recharge and refresh. Just be sure to set an alarm so you don't oversleep.

6. **Watch a funny movie or TV show**: Laughter is great medicine for stress and anxiety. Choose a lighthearted movie or TV show that makes you smile.

7. **Enjoy a warm cup of tea or coffee**: Brew yourself a warm cup of tea or coffee, and savor the flavors. You can also try herbal teas like peppermint or chamomile for a calming effect.

8. **Meditate or practice deep breathing exercises**: Take a few minutes to sit quietly, focus on your breath, and calm your mind. You can also try guided meditations or deep breathing exercises online.

9. **Take a walk or do some light exercise**: Get some fresh air and move your body with a short walk or some light stretching exercises. This can help release endorphins and clear your mind.

10. **Practice gratitude**: Take a few minutes to reflect on the things you're grateful for in your life. This can help shift your focus away from stress and anxiety.

11. **Get a massage**: Treat yourself to a professional massage or ask a partner or friend to give you a gentle massage at home.

12. **Play with a pet**: If you have a furry friend at home, spend some time playing with them. This can help reduce stress and increase feelings of relaxation.

13. **Write in a journal**: Write down your thoughts, feelings, and experiences. This can help you process your emotions and gain a new perspective.

14. **Do some creative activities**: Engage in a creative activity that brings you joy, such as drawing, painting, or knitting.

15. **Practice mindfulness**: Focus on the present moment and let go of worr
ies about the past or future. You can try mindfulness apps or guided meditat
ions online.

Remember, relaxation is a personal experience, and what works for one person
may not work for another. Experiment with different activities to find what
helps you feel most relaxed and rejuvenated.
Response 3: Relaxing after a long day is essential for rejuvenating your min
d, body, and soul. Here are some popular activities to help you unwind:

1. **Meditation and Yoga**: Practice mindfulness and meditation to calm your
mind and stretch your body. You can find guided sessions on YouTube or downl
oad meditation apps like Headspace or Calm.

2. **Reading**: Get lost in a book or a magazine that interests you. Reading
can help distract you from the stresses of the day and provide a much-needed
break.

3. **Listening to Music**: Put on your favorite playlist or discover new art
ists on music streaming platforms like Spotify or Apple Music. Let the sooth
ing melodies calm your mind.

4. **Taking a Warm Bath**: Fill your bathtub with warm water, add some relax
ing essential oils (like lavender or chamomile), and soak for a while. The w
arm water can help relax your muscles and calm your mind.

5. **Watching a Movie or TV Show**: Pick a movie or TV show you've been mean
ing to watch and get lost in a different world. You can also try binge-watch
ing a new series or exploring a new genre.

6. **Playing with Pets**: If you have a furry friend at home, spend some qua
lity time playing with them. Playing with pets can help reduce stress and pr
omote bonding.

7. **Taking a Short Walk**: Get some fresh air and take a short walk around
your neighborhood or a nearby park. Fresh air and nature can help clear your
mind and boost your mood.

8. **Sipping Herbal Tea**: Try herbal teas like chamomile, peppermint, or gi
nger, which can help calm your mind and promote relaxation.

9. **Playing a Board Game or Puzzle**: Engage your brain and challenge yours
elf with a board game or a puzzle. This can help distract you from the stres
ses of the day and provide a fun way to unwind.

10. **Practicing Deep Breathing**: Take a few minutes to focus on deep, slow
breaths, inhaling through your nose and exhaling through your mouth. This ca
n help calm your mind and reduce stress.

Remember, everyone is unique, so experiment with different activities to fin
d what works best for you.
Response 4: After a long day, it's essential to unwind and recharge. Here ar
e some great activities to help you relax:

1. **Reading**: Get lost in a book, magazine, or even a comic book. Reading
can help distract you from the stresses of the day and transport you to a di

fferent world.

2. **Meditation and Yoga**: Practice mindfulness, deep breathing, or simple stretches to calm your mind and body. You can find plenty of guided meditations and yoga routines online.

3. **Listening to Music**: Put on your favorite playlist, create a relaxing atmosphere, and let the soothing sounds calm your mind.

4. **Taking a Bath or Shower**: A warm bath or shower can be incredibly relaxing. Add some calming essential oils like lavender or chamomile to enhance the experience.

5. **Watching a Movie or TV Show**: Pick a calming show or a relaxing movie that helps you unwind. You can also try watching a documentary or a nature documentary.

6. **Playing with Pets**: If you have a furry friend, spend some quality time with them. Playing with pets can be a great way to relax and release endorphins.

7. **Gardening**: If you have a garden or a green thumb, spend some time tending to your plants. Gardening can be calming and fulfilling.

8. **Cooking or Baking**: Try out a new recipe or experiment with new flavors and ingredients. Cooking and baking can be therapeutic and enjoyable.

9. **Taking a Walk or Going for a Run**: Get some fresh air and take a walk or go for a run. Exercise can help release endorphins, which can improve your mood.

10. **Practicing Gratitude**: Take a few minutes to reflect on the things you're grateful for. Writing in a gratitude journal can be a great way to cultivate a positive mindset.

11. **Doing a Puzzle or Playing a Game**: Engage your brain with a crossword, Sudoku, or a board game. Solving a puzzle or playing a game can be a fun and relaxing activity.

12. **Getting a Massage**: Treat yourself to a professional massage or ask a partner or friend to give you a massage. Physical touch can be incredibly relaxing.

13. **Writing in a Journal**: Write down your thoughts, feelings, and experiences. Journaling can be a great way to process your emotions and reflect on your day.

14. **Sipping Herbal Tea**: Enjoy a cup of herbal tea, such as chamomile or peppermint, which can promote relaxation and calmness.

15. **Listening to ASMR**: Autonomous Sensory Meridian Response (ASMR) is a tingling sensation that some people experience when they listen to soft sounds, whispers, or other gentle stimuli. You can find plenty of ASMR content online.

Remember, relaxation is a personal experience, so experiment with different

activities to find what works best for you.
Response 5: There are many activities to help you relax after a long day. He
re are some suggestions:

1. **Meditation and deep breathing exercises**: Start by taking a few minute
s to sit comfortably, close your eyes, and focus on your breath. This helps
calm your mind and reduces stress.

2. **Yoga**: Engage in some gentle yoga stretches, poses, or even follow alo
ng with a video online. This can help loosen up your muscles, reduce tensio
n, and promote relaxation.

3. **Listen to soothing music**: Put on some calming music, nature sounds, o
r white noise to create a peaceful atmosphere. You can find many free resour
ces online, including guided meditations and relaxing playlists.

4. **Take a warm bath or shower**: Fill your bathtub with warm water, add so
me calming essential oils (like lavender or chamomile), and soak for a whil
e. Alternatively, take a warm shower to relax your muscles.

5. **Read a book**: Get lost in a good book, whether it's a fiction novel, a
self-help book, or a magazine. Reading can help distract you from the stress
es of the day and transport you to a different world.

6. **Watch a funny movie or TV show**: Laughter is the best medicine, after
all! Choose a lighthearted film or a comedy special to lift your mood and re
lax your mind.

7. **Get some fresh air**: Take a short walk outside, or open a window to le
t in some fresh air. This can help clear your mind and reduce stress.

8. **Do some light stretching**: Gentle stretching can help release tension
and relax your muscles. Try some simple stretches, like neck rolls, shoulder
rolls, or toe curls.

9. **Practice gratitude**: Reflect on the things you're grateful for, no mat
ter how small they may seem. Focusing on the positive can help shift your pe
rspective and calm your mind.

10. **Take a power nap**: If you're feeling exhausted, a 15-20 minute power
nap can be just what you need to recharge. Just be sure to set an alarm so y
ou don't oversleep!

11. **Play with a pet**: If you have a furry friend at home, spend some qual
ity time playing with them. Interacting with animals can be calming and soot
hing.

12. **Do a puzzle or play a game**: Engage your brain with a crossword, Sudo
ku, or a board game. Solving a puzzle or playing a game can be a fun way to
unwind and relax.

13. **Write in a journal**: Reflect on your day, writing down your thoughts
and feelings. This can help process your emotions and clear your mind.

14. **Practice progressive muscle relaxation**: Tense and then relax differe
nt muscle groups in your body, starting with your toes and moving up to your

head. This can help release physical tension.

15. **Get a massage**: Treat yourself to a professional massage or ask a partner to give you a gentle massage. This can help release tension and promote relaxation.

Remember, everyone is unique, so experiment with different activities to find what works best for you.

** The most consistent answer: Relaxing after a long day can be a great way to unwind and recharge. Here are some activities that can help you relax:

1. **Reading**: Get lost in a book, whether it's a fiction novel, a self-help book, or a magazine. Reading can be a great way to escape reality and relax.

2. **Listening to music**: Put on your favorite playlist or find a relaxing music station on streaming services. Singing along or simply enjoying the calming sounds can be soothing.

3. **Taking a warm bath or shower**: Fill your bathtub with warm water, add some bubbles or essential oils, and soak for a while. Alternatively, take a relaxing shower to help your muscles unwind.

4. **Meditation and yoga**: Practice mindfulness meditation, yoga, or guided relaxation exercises to calm your mind and body. You can find many free resources online.

5. **Taking a walk**: Get some fresh air and take a short walk around your neighborhood or a nearby park. Nature can be a great stress-reliever.

6. **Watching a movie or TV show**: Pick a favorite film or series and spend some time binge-watching. Just be sure to take breaks and relax before getting back to the screen.

7. **Playing with a pet**: If you have a furry friend at home, spend some time playing with them. Interacting with animals can be calming and stress-reducing.

8. **Cooking or baking**: Prepare a simple meal or bake some cookies. Cooking can be a fun way to relax and express your creativity.

9. **Doing some light stretching**: Take a few minutes to stretch your body, focusing on your major muscle groups. This can help release tension and relax your muscles.

10. **Practicing deep breathing**: Take slow, deep breaths, focusing on the sensation of the air moving in and out of your body. This can help calm your mind and reduce stress.

11. **Sipping herbal tea**: Try a calming herbal tea, such as chamomile or lavender, to help you relax and unwind.

12. **Playing a board game or card game**: Spend some time playing a low-key game, like solitaire or a puzzle, to relax and have fun.

13. **Writing in a journal**: Write down your thoughts, feelings, and experi
ences. Reflecting on your day can help you process your emotions and relax.

14. **Getting a massage**: Treat yourself to a professional massage or ask a
partner or friend to give you a massage at home. This can be a great way to
relax and unwind.

15. **Taking a power nap**: If you're feeling extremely tired, take a short
nap (20-30 minutes) to recharge and refresh your mind.

Remember, everyone is different, so experiment with different activities to
find what works best for you and relaxes you.

In [21]:
```
# generated knowledge prompting
question = "Why do planets orbit the sun?"

# generated knowledge steps
knowledge_prompts = [
    "Explain the concept of gravitational force.",
    "Describe how gravity affects objects in space.",
    "Explain how an object's mass influences the gravitational force it ex
    "Describe how gravitational force affects planetary motion around the
]

# collect knowledge statements
knowledge_statements = []
for k_prompt in knowledge_prompts:
    messages = [{"role": "user", "content": k_prompt}]
    response = Model(messages)[0]['generated_text'][1]['content']
    knowledge_statements.append(response)

# combine the knowledge statements to answer the main question
combined_knowledge = " ".join(knowledge_statements)
final_prompt = f"{combined_knowledge} Now, based on this information, why

# generate the final response using the accumulated knowledge
messages = [{"role": "user", "content": final_prompt}]
final_response = Model(messages)[0]['generated_text'][1]['content']

print("\n  ** Accumulated Knowledge Statements:")
for idx, statement in enumerate(knowledge_statements, start=1):
    print(f"Statement {idx}: {statement}")

print("\n  ** Final Answer:", final_response)
```

** Accumulated Knowledge Statements:
Statement 1: Gravitational force is a fundamental concept in physics that de
scribes the interaction between two objects with mass. It is one of the four
fundamental forces of nature, along with electromagnetism, the strong nuclea
r force, and the weak nuclear force.

The gravitational force is a result of the curvature of spacetime caused by
the presence of mass and energy. According to Albert Einstein's theory of ge
neral relativity, every object with mass warps the fabric of spacetime aroun
d it, creating a gravitational field.

Imagine a heavy bowling ball on a trampoline. The trampoline will warp and c
urve under the ball's weight, creating a depression in the center. This depr
ession is a perfect representation of the gravitational field around the bow
ling ball. The same principle applies to massive objects, such as planets, s
tars, and galaxies.

The strength of the gravitational force between two objects depends on the m
asses of the objects, the distance between them, and the direction of their
motion. The gravitational force is always attractive and is always directed
towards the center of the other object.

The formula for the gravitational force between two objects is:

F = G * (m1 * m2) / r^2

Where:

* F is the gravitational force
* G is the gravitational constant (6.67408e-11 N*m^2/kg^2)
* m1 and m2 are the masses of the objects
* r is the distance between the centers of the objects

For example, if you have two objects with masses of 5 kg and 10 kg, and they
are 10 meters apart, the gravitational force between them would be:

F = G * (5 kg * 10 kg) / (10 m)^2
F = 6.67408e-11 N*m^2/kg^2 * (50 kg^2)
F = 3.33528e-7 N

This is an incredibly small force, but it is still noticeable in many everyd
ay situations. For example, the force of gravity that pulls you towards the
center of the Earth is approximately 9.8 m/s^2.

In summary, gravitational force is a fundamental concept that describes the
interaction between massive objects. It is a result of the curvature of spac
etime caused by the presence of mass and energy. The strength of the gravita
tional force depends on the masses of the objects, the distance between the
m, and the direction of their motion.
Statement 2: Gravity is a fundamental force of nature that affects objects i
n space in unique ways. Here's how it impacts objects:

Gravitational Attraction: Gravity is the attractive force between two ob
jects with mass. According to Newton's law of universal gravitation, every p
oint mass attracts every other point mass with a force proportional to the p
roduct of their masses and inversely proportional to the square of the dista

nce between them.

Effects on Objects in Space:

1. **Orbital Motion:** Objects in orbit around a celestial body, such as the Moon or a planet, are constantly falling towards the center of the body due to gravity. This is known as orbital motion. The object's velocity is constantly increasing, and it begins to orbit the body at a stable distance.
2. **Gravitational Binding:** The force of gravity holds objects together in a state of gravitational binding. This means that objects will continue to attract each other unless they are separated by a significant distance.
3. **Escape Velocity:** Objects in space must have sufficient energy to escape the gravitational pull of a celestial body. The minimum energy required for an object to escape is known as the escape velocity.
4. **Gravitational Lensing:** Gravity can also bend the path of light passing near a massive object, a phenomenon known as gravitational lensing. This can cause the light from distant objects to be magnified or distorted.
5. **Gravitational Redshift:** The stronger the gravitational field, the more massive the object, and the greater the redshift (stretching of light) experienced by light emitted from the object.

Unique Effects in Space:

1. **Gravitational Time Dilation:** Objects in strong gravitational fields, such as near black holes or neutron stars, experience time dilation. Time passes slower for an observer near these objects due to the intense gravitational field.
2. **Gravitational Waves:** The acceleration of massive objects in strong gravitational fields can produce gravitational waves, ripples in the fabric of spacetime. These waves can be detected by spacecraft and other instruments.

In summary, gravity is a powerful force that affects objects in space in various ways, from orbital motion to gravitational binding and escape velocity. The unique effects of gravity in space, such as gravitational time dilation and gravitational waves, make it a fascinating area of study for scientists and engineers.
Statement 3: The mass of an object directly influences the gravitational force it exerts on other objects. The more massive an object, the stronger its gravitational pull. This is known as Newton's law of universal gravitation.

According to Newton's law, the gravitational force between two objects is proportional to the product of their masses and inversely proportional to the square of the distance between them. Mathematically, this can be represented as:

F = G * (m1 * m2) / r^2

Where:

* F is the gravitational force between the two objects
* G is the gravitational constant (6.67408e-11 N m^2 kg^-2)
* m1 and m2 are the masses of the two objects
* r is the distance between the centers of the two objects

The formula shows that as the mass of the object increases, the gravitational force it exerts on another object decreases. Conversely, as the distance b

etween the objects increases, the gravitational force decreases.

To illustrate this, consider the following examples:

* A small rock has a mass of 10 kg and is placed near a larger rock with a m
ass of 100 kg. The gravitational force between the two rocks is stronger bec
ause the larger rock has a greater mass.
* A person has a mass of 70 kg and is standing on the surface of the Earth,
which has a mass of approximately 5.972e24 kg. The gravitational force exert
ed by the Earth on the person is stronger because the Earth's mass is greate
r.

In summary, the mass of an object directly affects the strength of its gravi
tational pull on other objects. The more massive an object, the stronger its
gravitational force.
Statement 4: Gravitational force plays a crucial role in the motion of plane
ts around the sun. Here's a detailed explanation:

Gravitational Force Equation:

The gravitational force between two objects is described by the Universal La
w of Gravitation, which states that every point mass attracts every other po
int mass with a force that is directly proportional to the product of their
masses and inversely proportional to the square of the distance between thei
r centers.

F = G * (m1 * m2) / r^2

Where:
F = gravitational force
G = gravitational constant (6.67408e-11 N m^2 kg^-2)
m1 and m2 = masses of the objects
r = distance between the centers of the objects

Planetary Motion:

When a planet orbits the sun, it experiences a constant force of gravity, wh
ich keeps it in a stable, elliptical path. This force is responsible for the
planet's orbit being curved, rather than being a flat plane.

The gravitational force of the sun acts on the planet, causing it to move al
ong its orbit. The force is directed towards the center of the sun and is pr
oportional to the mass of the planet and the distance between the planet and
the sun.

Key Factors Affecting Planetary Motion:

Several factors influence the gravitational force on a planet's orbit:

1. **Mass of the sun:** The stronger the sun, the more gravitational force i
t exerts on a planet. This means that planets with a larger sun will have a
stronger gravitational pull.
2. **Radius of the orbit:** The distance between the planet and the sun affe
cts the strength of the gravitational force. Planets that orbit closer to th
e sun experience a stronger gravitational force.
3. **Orbital eccentricity:** The shape of the planet's orbit can affect the

gravitational force. Elliptical orbits are more common in our solar system, as they allow planets to maintain a stable balance between the sun's gravitational force and their own orbital velocity.

Effects of Gravitational Force on Planetary Motion:

The gravitational force of the sun affects the following aspects of planetary motion:

1. **Orbital period:** The time it takes a planet to complete one orbit around the sun is directly related to the strength of the sun's gravitational force.
2. **Orbital velocity:** The speed at which a planet moves around the sun is also influenced by the sun's gravitational force.
3. **Stability of the orbit:** The gravitational force helps maintain the stability of a planet's orbit, preventing it from becoming too eccentric or unstable.

In summary, the gravitational force of the sun plays a crucial role in the motion of planets around the sun. The strength of the sun's gravitational force determines the orbital period, orbital velocity, and stability of the planet's orbit, ultimately shaping the characteristics of our solar system.

** Final Answer: Planets orbit the sun because of the gravitational force exerted by the sun on the planets. Here's a simplified explanation of why planets orbit the sun:

Why Planets Orbit the Sun:

1. **Gravity:** The sun has a massive amount of energy, which is released as light and heat. This energy is transferred to the planets, causing them to move in elliptical orbits around the sun.
2. **Gravitational Force:** The sun's gravity pulls on the planets, causing them to move in a curved path around the sun. This force is stronger at the sun's center, resulting in a stronger gravitational pull.
3. **Conservation of Energy:** According to the law of conservation of energy, the total energy of a closed system remains constant. In this case, the energy from the sun's gravitational force is transferred to the planets, causing them to orbit the sun.

Why Planets Keep Their Orbits:

1. **Stability of the Orbit:** The gravitational force of the sun helps maintain the stability of the planet's orbit, preventing it from becoming too eccentric or unstable.
2. **Eccentricity:** The elliptical shape of a planet's orbit allows it to maintain a stable balance between the sun's gravitational force and its own orbital velocity.
3. **Conservation of Angular Momentum:** The conservation of angular momentum ensures that the planet's orbital velocity remains constant, allowing it to maintain its orbit around the sun.

Planetary Motion and the Gravitational Force:

The gravitational force of the sun affects the following aspects of planetary motion:

```
1. **Orbital Period:** The time it takes a planet to complete one orbit arou
nd the sun is directly related to the strength of the sun's gravitational fo
rce.
2. **Orbital Velocity:** The speed at which a planet moves around the sun is
also influenced by the sun's gravitational force.
3. **Stability of the Orbit:** The gravitational force helps maintain the st
ability of a planet's orbit, preventing it from becoming too eccentric or un
stable.

In summary, the gravitational force of the sun plays a crucial role in the m
otion of planets around the sun. The strength of the sun's gravitational for
ce determines the orbital period, orbital velocity, and stability of the pla
net's orbit, ultimately shaping the characteristics of our solar system.
```

Key Features of LLaMA **Instruct** Models

LLaMA (Large Language Model Meta AI) is a family of foundational models created by Meta. While the original LLaMA models are designed for general-purpose language understanding and generation tasks, LLaMA Instruct models are fine-tuned versions that are optimized for instruction-following tasks, focusing on better understanding and responding to human instructions.

- **Instruction-fine-tuned** on datasets specifically curated for instruction-following tasks
  - Human-provided instructions and their responses, e.g., Q-n-A
  - Artificially generated instructions paired with high-quality responses
  - Feedback loops that enhance the model's ability to provide concise and accurate answers
  - Fine-tuning to have the model align better with human intentions and prioritize direct and **task-specific responses** over verbose or irrelevant outputs
- Enhanced dialogue understanding, which handles multi-turn dialogues more effectively
  - Maintain context across turns
  - Provide coherent and contextually relevant answers
  - Ask clarifying questions when user instructions are ambiguous
- Safety and alignment by training on instruction-focused datasets, these models are less likely to generate harmful, misleading, or biased outputs
  - Filtering datasets to reduce exposure to toxic content
  - Reinforcement learning with human feedback (RLHF) to align model behavior with user expectations
  - The alignment makes them more suitable for deployment in consumer-facing applications
- Adaptability to use cases
  - Education: Answering queries in a structured, step-by-step manner

- Productivity tools: Summarizing, translating, or explaining content
- Coding: Debugging, writing scripts, or explaining programming concepts
- Research: Providing concise and reliable explanations for complex topics

LLaMA Instruct is comparable to other instruction-tuned models, such as OpenAI's InstructGPT and Google's T5-based models. Key differences include the following table,

Feature	LLaMA Instruct	InstructGPT	T5/Flan Models
Training Data	Open-sourced datasets with a focus on alignment	Proprietary datasets curated by OpenAI	Public datasets with diverse instructions
Deployment	Open for research and non-commercial use	Commercially available via OpenAI API	Open-source or publicly available
Fine-tuning	RLHF or instruction-specific datasets	RLHF extensively used	Mixture of fine-tuning methods
Performance on Tasks	Excels in understanding and generating human-like responses in open-ended tasks	State-of-the-art on multiple instruction tasks	Good generalist for NLP tasks

In [22]:
```
# Try the 3B model
Model = pipeline("text-generation", model="meta-llama/Llama-3.2-3B-Instruc
```

Loading checkpoint shards: 100% 2/2 [00:00<00:00, 24.38it/s]

In [23]:
```
Markdown(query(Model, 'The sky is ...'))
```

Out[23]: ...blue. Would you like to complete the sentence or talk about something else?

In [24]:
```
# generated knowledge prompting
question = "Why do planets orbit the sun?"

# generated knowledge steps
knowledge_prompts = [
    "Explain the concept of gravitational force.",
    "Describe how gravity affects objects in space.",
    "Explain how an object's mass influences the gravitational force it ex
    "Describe how gravitational force affects planetary motion around the
]

# collect knowledge statements
knowledge_statements = []
for k_prompt in knowledge_prompts:
    messages = [{"role": "user", "content": k_prompt}]
    response = Model(messages)[0]['generated_text'][1]['content']
    knowledge_statements.append(response)

# combine the knowledge statements to answer the main question
```

```python
combined_knowledge = " ".join(knowledge_statements)
final_prompt = f"{combined_knowledge} Now, based on this information, why

# generate the final response using the accumulated knowledge
messages = [{"role": "user", "content": final_prompt}]
final_response = Model(messages)[0]['generated_text'][1]['content']

print("\n  ** Accumulated Knowledge Statements:")
for idx, statement in enumerate(knowledge_statements, start=1):
    print(f"Statement {idx}: {statement}")

print("\n  ** Final Answer:", final_response)
```

** Accumulated Knowledge Statements:
Statement 1: The concept of gravitational force is a fundamental principle in physics that describes the attractive force between two objects with mass. It is a universal force that acts between all objects with mass, from the smallest subatomic particles to the largest galaxies.

History and Discovery

The concept of gravitational force was first described by Sir Isaac Newton in his groundbreaking work, "Philosophiæ Naturalis Principia Mathematica" (Mathematical Principles of Natural Philosophy) in 1687. Newton's law of universal gravitation states that every point mass attracts every other point mass by a force acting along the line intersecting both points.

Key Principles

The gravitational force between two objects is determined by the following factors:

1. **Mass**: The more massive the objects, the stronger the gravitational force between them.
2. **Distance**: The gravitational force decreases with increasing distance between the objects.
3. **Inverse Square Law**: The gravitational force is inversely proportional to the square of the distance between the objects.

Mathematically, the gravitational force (F) between two objects with mass (m1 and m2) is described by the following equation:

F = G * (m1 * m2) / r^2

where:

* F is the gravitational force
* G is the gravitational constant (6.67408e-11 N*m^2/kg^2)
* m1 and m2 are the masses of the two objects
* r is the distance between the centers of the two objects

Types of Gravitational Forces

There are two types of gravitational forces:

1. **Attractive Gravitational Force**: The force that attracts two objects towards each other.
2. **Repulsive Gravitational Force**: The force that pushes two objects apart. However, this type of force is extremely weak and only exists in certain situations, such as the repulsion between two objects with the same sign of mass.

Gravitational Effects

The gravitational force has significant effects on the behavior of objects in the universe, including:

1. **Orbiting**: The gravitational force holds planets in orbit around their stars.

2. **Gravity Wells**: The gravitational force warps the fabric of spacetime, creating a gravitational well that affects the motion of objects.
3. **Gravitational Waves**: The gravitational force produces ripples in spac etime that can be detected and studied.

In summary, the concept of gravitational force is a fundamental principle in physics that describes the attractive force between two objects with mass. T he gravitational force is determined by the masses of the objects and the di stance between them, and it has significant effects on the behavior of objec ts in the universe.
Statement 2: In space, gravity affects objects in a unique way compared to E arth's surface. Here's a breakdown of how gravity behaves in space:

1. **Gravity is weaker in space**: Since space is a vacuum, there is no air resistance or friction to slow down objects. As a result, objects in space c an move faster and farther than they would on Earth. The strength of gravity also decreases with distance from a celestial body, so the farther away an o bject is from the center of a planet or star, the weaker the gravitational p ull.

2. **Gravity affects orbiting objects**: In space, objects in orbit around a celestial body follow elliptical paths due to the gravitational pull. The fo rce of gravity keeps the object moving in a curved path, but it's not strong enough to pull the object back down to the surface. This is why satellites, planets, and stars orbit around each other.

3. **Gravity warps space-time**: According to Einstein's theory of general r elativity, gravity is not just a force that pulls objects towards each othe r, but also warps the fabric of space-time around massive objects. This mean s that the curvature of space-time affects the motion of objects in space, c ausing them to follow curved paths.

4. **Gravity affects the motion of comets and asteroids**: Comets and astero ids are objects that are in orbit around the Sun. As they approach the Sun, the increased gravity causes them to slow down and eventually fall towards t he Sun due to the strong gravitational force.

5. **Gravity affects the formation of galaxies**: Gravity plays a crucial ro le in the formation and evolution of galaxies. Galaxies are held together by gravity, and the force of gravity helps to shape the distribution of stars a nd other objects within them.

6. **Gravity affects the motion of black holes**: Black holes are regions of space where the gravitational pull is so strong that nothing, including ligh t, can escape. The gravity of a black hole warps space-time in extreme ways, causing objects to move along curved paths that are influenced by the black hole's mass and spin.

In summary, gravity plays a vital role in shaping the behavior of objects in space, from the orbits of planets and satellites to the formation of galaxie s and the motion of black holes.
Statement 3: An object's mass plays a crucial role in determining the gravit ational force it exerts on other objects. According to Newton's Law of Unive rsal Gravitation, the gravitational force (F) between two objects is directl y proportional to the product of their masses (m1 and m2) and inversely prop ortional to the square of the distance (r) between their centers.

Mathematically, this is represented by the equation:

F = G * (m1 * m2) / r^2

where:

- F is the gravitational force between the two objects
- G is the gravitational constant (approximately 6.67408e-11 N*m^2/kg^2)
- m1 and m2 are the masses of the two objects
- r is the distance between the centers of the two objects

As you can see from the equation, the gravitational force is directly proportional to the product of the masses. This means that if you double the mass of one object, the gravitational force it exerts on another object will also double.

For example, if you have two objects with masses of 10 kg and 20 kg, and the distance between them is 1 meter, the gravitational force they exert on each other is:

F = G * (10 kg * 20 kg) / (1 m)^2
= G * 200 kg^2 / 1 m^2
= 1.3464e-9 N

Now, if you double the mass of one of the objects to 40 kg, the gravitational force they exert on each other becomes:

F = G * (10 kg * 40 kg) / (1 m)^2
= G * 400 kg^2 / 1 m^2
= 5.3728e-9 N

As you can see, the gravitational force has doubled, even though the distance between the objects remains the same.

In summary, an object's mass directly influences the gravitational force it exerts on other objects. The more massive an object is, the stronger the gravitational force it exerts on other objects.
Statement 4: Gravitational force plays a crucial role in the motion of planets around the Sun. According to Newton's law of universal gravitation, every object in the universe attracts every other object with a force proportional to their masses and inversely proportional to the square of the distance between them.

In the case of planetary motion, the gravitational force exerted by the Sun on a planet is what keeps it in orbit around the Sun. The force is directed towards the center of the Sun, which is why planets follow curved paths, or orbits, around the Sun.

Here are some key ways in which gravitational force affects planetary motion:

1. **Centripetal force**: The gravitational force acting on a planet is the centripetal force that keeps it in orbit around the Sun. This force is responsible for the curvature of the planet's path and prevents it from flying off into space.

2. **Orbital velocity**: The gravitational force also determines the orbital velocity of a planet. The faster a planet moves, the stronger the gravitational force acting on it, and the more it is attracted to the Sun.

3. **Orbital period**: The gravitational force affects the orbital period of a planet, which is the time it takes for the planet to complete one orbit around the Sun. The longer the orbital period, the weaker the gravitational force acting on the planet.

4. **Eccentricity**: The gravitational force also affects the eccentricity of a planet's orbit. Eccentricity is a measure of how elliptical the orbit is. A highly eccentric orbit means that the planet is closer to the Sun at one point and farther away at another.

5. **Stable orbits**: The gravitational force ensures that planets maintain stable orbits around the Sun. If a planet were to suddenly lose its gravitational connection to the Sun, it would either fly off into space or be pulled into a smaller orbit.

In summary, the gravitational force exerted by the Sun on a planet is what keeps it in orbit around the Sun, determines its orbital velocity, orbital period, and eccentricity, and ensures the stability of the planet's orbit.

** Final Answer: The concept of gravitational force is a fundamental principle in physics that describes the attractive force between two objects with mass. It is a universal force that acts between all objects with mass, from the smallest subatomic particles to the largest galaxies.

History and Discovery

The concept of gravitational force was first described by Sir Isaac Newton in his groundbreaking work, "Philosophiæ Naturalis Principia Mathematica" (Mathematical Principles of Natural Philosophy) in 1687. Newton's law of universal gravitation states that every point mass attracts every other point mass by a force acting along the line intersecting both points.

Key Principles

The gravitational force between two objects is determined by the following factors:

1. **Mass**: The more massive the objects, the stronger the gravitational force between them.
2. **Distance**: The gravitational force decreases with increasing distance between the objects.
3. **Inverse Square Law**: The gravitational force is inversely proportional to the square of the distance between the objects.

Mathematically, the gravitational force (F) between two objects with mass (m1 and m2) is described by the following equation:

$$F = G * (m1 * m2) / r^2$$

where:

* F is the gravitational force
* G is the gravitational constant (6.67408e-11 N*m^2/kg^2)
* m1 and m2 are the masses of the two objects
* r is the distance between the centers of the two objects

Types of Gravitational Forces

There are two types of gravitational forces:

1. **Attractive Gravitational Force**: The force that attracts two objects towards each other.
2. **Repulsive Gravitational Force**: The force that pushes two objects apart. However, this type of force is extremely weak and only exists in certain situations, such as the repulsion between two objects with the same sign of mass.

Gravitational Effects

The gravitational force has significant effects on the behavior of objects in the universe, including:

1. **Orbiting**: The gravitational force holds planets in orbit around their stars.
2. **Gravity Wells**: The gravitational force warps the fabric of spacetime, creating a gravitational well that affects the motion of objects.
3. **Gravitational Waves**: The gravitational force produces ripples in spacetime that can be detected and studied.

Gravity in Space

Gravity behaves differently in space compared to Earth's surface. Here are some key ways in which gravity affects objects in space:

1. **Gravity is weaker in space**: Since space is a vacuum, there is no air resistance or friction to slow down objects. As a result, objects in space can move faster and farther than they would on Earth. The strength of gravity also decreases with distance from a celestial body, so the farther away an object is from the center of a planet or star, the weaker the gravitational pull.

2. **Gravity affects orbiting objects**: In space, objects in orbit around a celestial body follow elliptical paths due to the gravitational pull. The force of gravity keeps the object moving in a curved path, but it's not strong enough to pull the object back down to the surface. This is why satellites, planets, and stars orbit around each other.

3. **Gravity warps space-time**: According to Einstein's theory of general relativity, gravity is not just a force that pulls objects towards each other, but also warps the fabric of space-time around massive objects. This means that the curvature of space-time affects the motion of objects in space, causing them to follow curved paths.

4. **Gravity affects the motion of comets and asteroids**: Comets and asteroids are objects that are in orbit around the Sun. As they approach the Sun, the increased gravity causes them to slow down and eventually fall towards the Sun due to the strong gravitational force.

5. **Gravity affects the formation of galaxies**: Gravity plays a crucial role in the formation and evolution of galaxies. Galaxies are held together by gravity, and the force of gravity helps to shape the distribution of stars and other objects within them.

6. **Gravity affects the motion of black holes**: Black holes are regions of space where the gravitational pull is so strong that nothing, including light, can escape. The gravity of a black hole warps space-time in extreme ways, causing objects to move along curved paths that are influenced by the black hole's mass and spin.

In summary, gravity plays a vital role in shaping the behavior of objects in space, from the orbits of planets and satellites to the formation of galaxies and the motion of black holes. An object's mass plays a crucial role in determining the gravitational force it exerts on other objects. According to Newton's Law of Universal Gravitation, the gravitational force (F) between two objects is directly proportional to the product of their masses (m1 and m2) and inversely proportional to the square of the distance (r) between their centers.

Mathematically, this is represented by the equation:

F = G * (m1 * m2) / r^2

where:

- F is the gravitational force between the two objects
- G is the gravitational constant (approximately 6.67408e-11 N*m^2/kg^2)
- m1 and m2 are the masses of the two objects
- r is the distance between the centers of the two objects

As you can see from the equation, the gravitational force is directly proportional to the product of the masses. This means that if you double the mass of one object, the gravitational force it exerts on another object will also double.

For example, if you have two objects with masses of 10 kg and 20 kg, and the distance between them is 1 meter, the gravitational force they exert on each other is:

F = G * (10 kg * 20 kg) / (1 m)^2
= G * 200 kg^2 / 1 m^2
= 1.3464e-9 N

Now, if you double the mass of one of the objects to 40 kg, the gravitational force they exert on each other becomes:

F = G * (10 kg * 40 kg) / (1 m)^2
= G * 400 kg^2 / 1 m^2
= 5.3728e-9 N

As you can see, the gravitational force has doubled, even though the distance between the objects remains the same.

In summary, an object's mass directly influences the gravitational force it

exerts on other objects. The more massive an object is, the stronger the gra
vitational force it exerts on other objects. Gravitational force plays a cru
cial role in the motion of planets around the Sun. According to Newton's law
of universal gravitation, every object in the universe attracts every other
object with a force proportional to their masses and inversely proportional
to the square of the distance between them.

In the case of planetary motion, the gravitational force exerted by the Sun
on a planet is what keeps it in orbit around the Sun. The force is directed
towards the center of the Sun, which is why planets follow curved paths, or
orbits, around the Sun.

Here are some key ways in which gravitational force affects planetary motio
n:

1. **Centripetal force**: The gravitational force acting on a planet is the
centripetal force that keeps it in orbit around the Sun. This force is respo
nsible for the curvature of the planet's path and prevents it from flying of
f into space.

2. **Orbital velocity**: The gravitational force also determines the orbital
velocity of a planet. The faster a planet moves, the stronger the gravitatio
nal force acting on it, and the more it is attracted to the Sun.

3. **Orbital period**: The gravitational force affects the orbital period of
a planet, which is the time it takes for the planet to complete one orbit ar
ound the Sun. The longer the orbital period, the weaker the gravitational fo
rce acting on the planet.

4. **Eccentricity**: The gravitational force also affects the eccentricity o
f a planet's orbit. Eccentricity is a measure of how elliptical the orbit i
s. A highly eccentric orbit means that the planet is closer to the Sun at on
e point and farther away at another.

5. **Stable orbits**: The gravitational force ensures that planets maintain
stable orbits around the Sun. If a planet were to suddenly lose its gravitat
ional connection to the Sun, it would either fly off into space or be pulled
into a smaller orbit.

In summary, the gravitational force exerted by the Sun on a planet is what k
eeps it in orbit around the Sun, determines its orbital velocity, orbital pe
riod, and eccentricity, and ensures the stability of the planet's orbit.

In [25]:
```
prompt= """Explain the philosophical implications of Gödel's incompletenes
referencing connections to quantum mechanics, the limits of human cognitio

Markdown(query(Model, prompt))
```

Out[25]: Gödel's incompleteness theorems imply that any formal system powerful enough to describe the universe must either be incomplete or inconsistent. This challenges the pursuit of a unified theory of everything in physics, as no complete and consistent theory can be proven within the system. Quantum mechanics, with its probabilistic nature, may be partially compatible, but the limits of human cognition and the nature of mathematical truth raise questions about the scope of a unified theory. Gödel's theorems suggest that true understanding may be limited by the inherent limitations of language and cognition, rather than the universe itself.

In [26]:
```
# This prompt is difficult because it requires the model to synthesize ide
# mathematics, theoretical physics, and cognitive science concisely,
# while also respecting word limits and maintaining logical coherence
```

In [27]:
```
%%html
<style>
    table {margin-left: 0 !important;}
    p {font-family: verdana;}
    li {font-family: verdana;}
    div {font-size: 10pt;}
</style>
<!-- Display markdown tables left oriented in this notebook. -->
```